

Data Structure (C Programming)

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Name : Harshad Sanjay Teli.

Roll No : 15

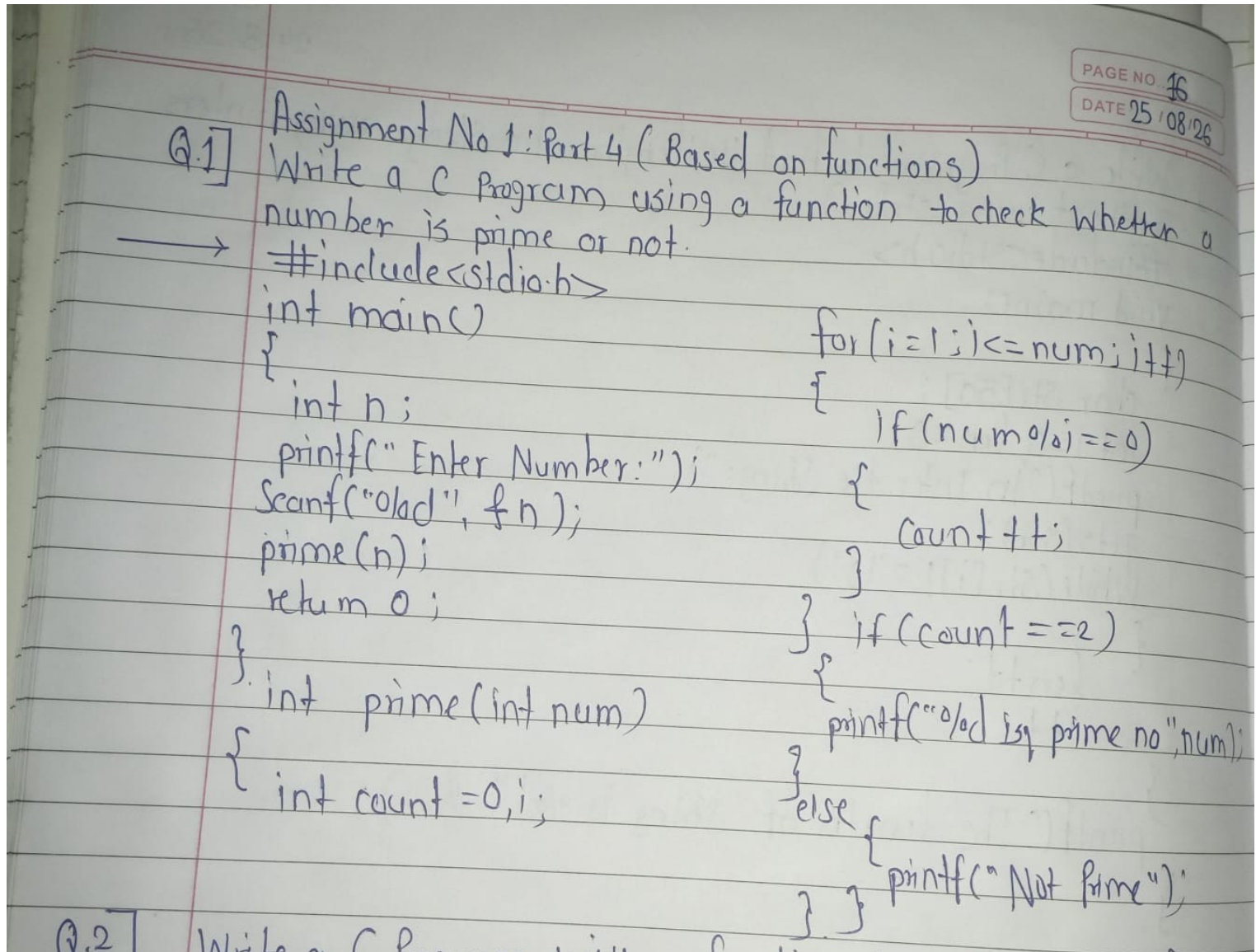
Batch : B1

Prerequisite

Based on Functions.

Q1. Write a C program using a function to check whether a number is prime or not.

Input:



Output:

```
C:\Users\Harshad Teli\Downloads\DS>output
Enter Number:7
7 is Prime Number
C:\Users\Harshad Teli\Downloads\DS>output
Enter Number:56
56 is Not Prime Number
C:\Users\Harshad Teli\Downloads\DS>|
```

Q2. Write a C program with a function to calculate the factorial of a number.

Input:

Q.2] Write a C Program with a function to calculate the factorial of a number.

```
→ #include <stdio.h>
int fact(int a);
int main()
{
    int a;
    printf("Enter Number:");
    scanf("%d", &a);
    int result = fact(a);
    printf("The factorial is %d", result);
    return 0;
}

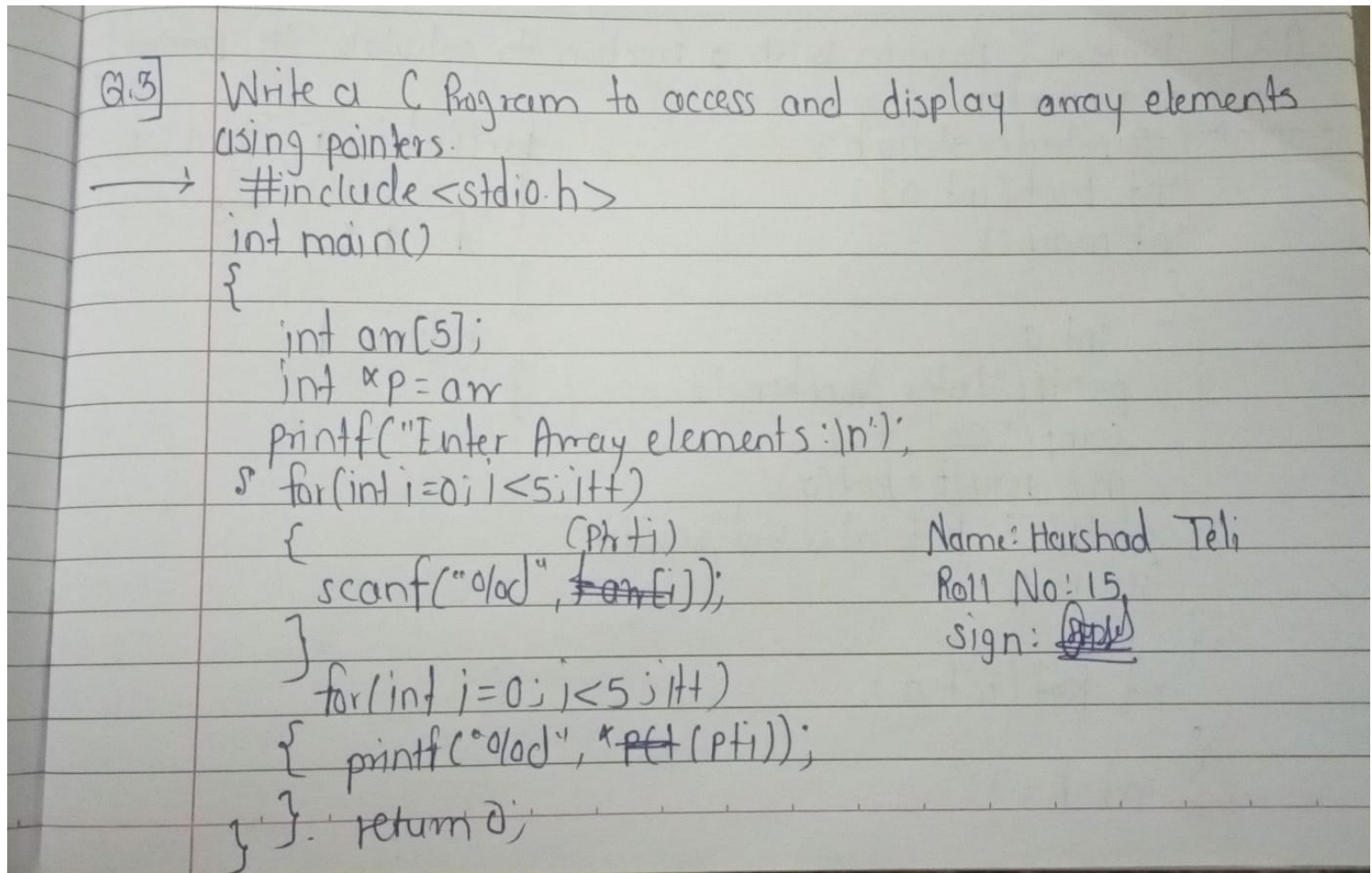
int fact(int a)
{
    int f = 1;
    for(int i = a; i >= 1; i--)
    {
        f = f * i;
    }
    return f;
}
```

Output:

```
The factorial of the 5 is 120
student@student-OptiPlex-SFF-7020:~/Pictures/Harshad$ gcc -o output pointer1.c
student@student-OptiPlex-SFF-7020:~/Pictures/Harshad$ ./output
Enter Number:6
The Factorial of the 6 is 720
student@student-OptiPlex-SFF-7020:~/Pictures/Harshad$
```

Q3. Write a C program with a function that takes two integers as parameters and returns their greatest common divisor (GCD).

Input:



Output:

